KMI ZEOLITE

GROWER PRODUCT GUIDE

High-Purity Natural Clinoptilolite Zeolite for Agriculture

What Is KMI Zeolite?

KMI Zeolite is a naturally mined clinoptilolite zeolite sourced from one of the largest and purest deposits in the United States, located near Pahrump, Nevada. For three generations, KMI has mined, milled, and distributed this high-purity mineral to growers and agricultural operations across North America and around the world.

Clinoptilolite zeolite forms when volcanic ash crystallizes over thousands of years in the presence of water, producing a honeycomb-like mineral lattice packed with millions of tiny pores and channels. This unique structure is what makes zeolite so valuable in the field — it holds, exchanges, and slowly releases nutrients and water exactly where crops need them.

OMRI Listed. KMI Zeolite is certified for use in certified organic production. It is also FDA-approved for use as an animal feed supplement and dietary mineral.

Why Zeolite Works in Agricultural Soils

The core mechanism behind zeolite's performance is its cation exchange capacity (CEC) — the ability to attract, hold, and release positively charged nutrient ions such as ammonium (NH_4^+), potassium (K^+), calcium (Ca^{2+}), and magnesium (Mg^{2+}). Rather than allowing these nutrients to leach out with irrigation or rainfall, zeolite acts as a reservoir, holding them in the root zone and releasing them as the plant demands.

Key mechanisms:

- High surface area (40 m^2/g) provides enormous contact area for nutrient and water interaction
- 50% pore volume means it holds significant amounts of water and gases within its structure
- Ion exchange captures ammonium from fertilizer applications, preventing volatilization and leaching
- Structural stability: does not break down under normal field conditions — one application provides lasting benefit

Soil & Crop Benefits

Nutrient Retention & Use Efficiency

Up to 70% of applied nitrogen can be lost to the environment through volatilization, leaching, and runoff under standard fertilizer programs. Zeolite significantly reduces these losses by capturing ammonium ions within its lattice and releasing them slowly as crops need them.

- Increases N, P, and K uptake efficiency in roots, leaves, and stems
- Reduces fertilizer application frequency and total input costs
- Keeps nutrients in the root zone — not in the water table

Water Retention & Irrigation Efficiency

Zeolite's porous structure acts as a sponge in the root zone, holding water between irrigation cycles and reducing the frequency and volume of water needed to maintain optimal soil moisture.

- Reduces water requirements, especially valuable in restricted-water regions
- Improves moisture distribution in sandy or well-drained soils
- Reduces the risk of drought stress between irrigations

Soil Structure & Aeration

- Prevents soil compaction in high-traffic or clay-heavy fields
- Increases infiltration rates — reduces surface runoff and erosion
- Improves aeration of deep root systems due to high porosity
- Supports beneficial microbial communities in the soil

Heavy Metal & Toxin Management

Zeolite has a demonstrated ability to encapsulate heavy metals and bind mycotoxins such as aflatoxins and ochratoxins. This makes it particularly valuable on soils with contamination history or in operations where commodity grain quality and aflatoxin compliance are concerns.

- Binds lead, cadmium, and other heavy metal cations
- Reduces aflatoxin and ochratoxin mobility in degraded soils
- Can help remediate contaminated or historically over-irrigated ground

Technical Specifications

KMI Zeolite is 97% pure clinoptilolite, independently verified by third-party laboratory analysis. The deposit maintains consistent CEC across many acres and hundreds of feet in depth — meaning the product you receive today performs the same as product from last season.

Property	Value
Mineral Type	Clinoptilolite-Ca (Natural Zeolite)
Purity	97% Clinoptilolite
Cation Exchange Capacity	2.0 mEq/g

Specific Surface Area	40 m²/g
Pore Volume	50%
Pore Size	4–7.0 Angstroms
pH Stability	3.0–10.0 (stable across agricultural pH range)
Hardness (Mohs)	4–5.0
Thermal Stability	Up to 700°C
Crush Strength	2,500 lb/in²
Bulk Weight (Granular)	~50 lbs per cubic foot

Application Rates & Methods

Because zeolite is a physical mineral amendment rather than a consumable input, rates vary widely depending on soil type, crop, and goals. The following are general starting points — soil testing before and after application is recommended.

Broadcast & Incorporate (Pre-Plant)

- General rate: 500–2,000 lbs/acre, incorporated 4–8 inches
- Sandy or low-CEC soils: err toward the higher end
- Heavy clay soils: lower rates focused on structure and drainage
- Apply before tillage for best soil incorporation

Fertilizer Blend Carrier

- Blend with dry fertilizer at 10–20% by weight to improve nutrient retention
- 14×30 or 30-Minus grades recommended for blending uniformity
- Compatible with most granular fertilizer blending systems

Compost & Green Waste

- Add at 5–10% by volume in composting operations
- Absorbs ammonia during active decomposition — reduces odor and nitrogen loss
- Finished compost retains zeolite benefit when applied to fields

Turf & Specialty Crops

- Golf courses and turf: 1–3 lbs/sq ft incorporated into root zone during renovation
- Orchards and vineyards: banded at transplanting or broadcast and disked
- Greenhouses and nurseries: 10–20% by volume in potting media

Persistence: Unlike fertilizers, zeolite does not deplete or break down under normal field conditions. A single application provides years of continued benefit, improving the ROI with each subsequent season.

Certifications & Compliance

Property	Value
OMRI Listed	Approved for certified organic production — multiple categories
FDA Status	Generally Recognized As Safe (GRAS) for all animal species
Toxicity	Non-toxic, non-hazardous, fully compostable
Environmental	Does not leach; environmentally inert; 100% natural mineral